Question	Answer	Marks
10(a)(i)	ultrasound waves make the <u>crystal</u> in the transducer vibrate	**B1**
	vibrations (of crystal) cause an <u>induced</u> e.m.f. (that is detected)	**B1**
10(a)(ii)	product of density and speed	**M1**
	speed of sound in medium (and density of medium)	**A1**

Question	Answer	Marks
10(b)(i)	speed $= (1.49 \times 10^6) / 998$	**C1**
	$=$ **1490 m s^{-1}**	**A1**
10(b)(ii)	coefficient $= [(9.98 - 1.49) / (9.98 + 1.49)]^2$	**C1**
	$=$ **0.548**	**A1**
10(b)(iii)	any **three** points from: • (specific acoustic) impedance values are very different • intensity reflection coefficient is (approximately) equal to 1 • (almost) all the sound (intensity) is reflected (at the boundary) • (virtually) none of the sound (intensity) is transmitted (into the glass)	**B3**